



ANALYSING THE AWARENESS OF CYBER CRIME AND DESIGNING A RELEVANT FRAMEWORK WITH RESPECT TO CYBER WARFARE: AN EMPIRICAL STUDY

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ABSTRACT

Internet users are progressively getting themselves vulnerable to security dangers amid their utilization of the internet. Therefore, the concern can be raised about users' awareness of these issues, and the degree to which they are thus secured and prepared to manage situations they may experience. This paper has provided a survey of contemporary thought on the challenges presented by cybercrime. It began by looking at existing definitions of cybercrime and cyberwarfare and found two problems that needed resolving. Firstly, it was found that there is no widely accepted definition of either cybercrime or cyber warfare. This is problematic since, without an agreed definition, it is difficult to discuss the deeper issues or even recognize when cyber warfare is occurring. It presents outcomes from the study of 325 users to evaluate their view of security issues, their awareness and their dispositions towards the utilization of related software. The findings uncover that in spite of the fact that there is a high level of confidence at a surface level, with respondents asserting to know about the dangers and using a large number of the important protections, more thorough interpretation proposes that there are a few regions in which essential information and comprehension are inadequate. Even though a large portion of the issues was typically intense among fledgeling users, there was additionally outstanding ignorance among users who viewed themselves to have propelled levels of internet knowledge.

Key words: internet users, Security Dangers, Security issues, Survey data, user problems.

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1. INTRODUCTION

As the Internet has developed, become more accessible and taken on greater importance in our societies, terrorists have logically followed the trend and increasingly used it. But the combination of this new medium and the growing threat of global crime calls traditional legal responses into question and shakes them to the core. States face three major difficulties in effectively responding to cyber-crime. The need to identify both criminals and territorial competence are greatly challenged by the loose and transnational structure of cyber-terrorist groups. The anonymity offered by the Internet shakes up both police investigations and the judicial principle of the individual nature of penalties. Similarly, the importance of understanding the methods and risks a criminal activity triggers is largely complicated by the highly technical nature of the devices and networks used, requiring specially trained experts in the fight against cyber-crime. The necessity of adequately responding to the terrorist cyber-threat is also defied by the rapidity of communications and the virtual nature of the information exchanged by terrorists over the Internet(Von Solms & Van Niekerk, 2013).

Furthermore, crime goes hand in hand with political demands or goals. Because liberal democracy is based on the allowance of free debate of political views, prosecutions based on the diffusion of terrorist messages or content on the Internet question the very core of our fundamental principles. The appropriate balance between civil liberties and the fight against assailants of our societies is hard to reach and always unstable. The first difficulty comes from the States' incapacity to reach an agreement on a universal definition of the crime, although a European definition has been adopted. Logically, "cyber-crime" as a concept is greatly discussed. Some scholars argue that it should be limited to cyber-attacks carried out by terrorists whereas others contend that it should encompass all uses of the Internet for terrorist purposes (Droege, 2013).

Throughout history, mankind has waged war, seeking to further national agendas in an ever-changing international game of power. From the sword battles of the past to the unmanned drone strikes of today, this game of power is constantly driven to shift and evolve by technology. The development of armoured vehicles, aircraft, ships and the use of electronics and telecommunications have all expanded the battle space and introduced new and innovative ways to gain an advantage over opponents. Just as the technological innovation of flight triggered a race to dominate the skies, the emergence of cyberspace has opened up new strategic possibilities and threats, causing a scramble to secure a dominant position inside of it(Droege, 2013; Wilson, 2014).

Governments, too, are fully aware of the need to take action in response to threats from cyberspace. US President Barack Obama has declared America's digital infrastructure a strategic national asset and formed Cybercom: a division inside the Pentagon whose stated task is to "perform full spectrum operations" (War in the fifth domain, 2010). Documents leaked from the National Security Agency in the US also confirm that national security figures are seeking to establish offensive cyber capability (Denning, 2014; Everett, 2015) (In the UK, government officials have warned of a lack of preparedness for cyber warfare and have announced new investments to bolster defense, such as the National Cyber Security

Programmed (Uk 'complacent' over military cyber-attack risk, mps warn, 2013). NATO has also been raising awareness, releasing the Tallinn Manual on the International Law Applicable to Cyber Warfare (NATO, 2013) as an attempt to advise nations on how to operate legally in this new warfighting domain. Looking at this evidence, it is clear that cyber warfare is a topic of global concern (Dogrul, Aslan, & Celik, 2011).

The advent of the Internet is the most important development of the modern day communication system but it has brought a new age of crime, termed as cybercrimes. The paper majorly focuses on the two common types of cybercrimes - Cyber Warfare and Cyber Terrorism, although the concept of the two appears to be similar. The paper, going into the insight of the cybercrime, cyber warfare and the interconnection between them.

2. LITERATURE REVIEW

2.1. Cybercrime

The dramatization of the scene of innovation now penetrating our regular day to day existences, joined with the dread of the possibly dystopic fates it could incur upon us ensures news articles about "cybercrime" a place on the front page. However, very frequently the cases made in those articles about the pervasiveness of cybercrimes need light in the matter of what it is that is especially "digital" about them. In reality, on the uncommon events that purported instances of cybercrime come to court – commonly involving Internet extortion, burglary, erotic entertainment, paedophilia, notwithstanding hacking, and so forth – they regularly have the natural ring of the "customary" instead of the "digital" about them (Team, Bodin, Echilley, & Quinard-Thibault, 2015).

The expression "cybercrime" is generally utilized today to depict the violations or damages that outcome from openings made by arranged advances (Kenneth, Pandemonium, States, & Security, n.d.). Its roots lie in sci-fi discussions, books and movies that logically connected "the internet" with "felony" to portray violations that occur in virtual situations. The term the internet, for instance, seems to have been instituted by William Gibson in his 1982 short story 'Consuming Chrome' distributed in Omni Magazine– a sci-fi meets science discussion that existed in the vicinity of 1978 and 1998. The idea of the internet and virtual situations then advanced through cyberpunk sci-fi short stories by Bruce Bethke and others, and books, for example, Gibson's Neuromancer and Stephenson's Snowcrash. The linkage amongst the internet and felony was constantly expressed in the idea of cyberpunk and it encircled most emotional fiction movies, however, the real term "cybercrime" appears to have risen in the late 1980s or even mid-1990s in the later cyberpunk books and comic books. The social combination of the internet and wrongdoing into the standard mainstream culture is to a great extent because of three eras of films depicting programmers which imbued a portion of the cyberpunk thoughts. The first is the early original movies characterized by 'the hack' into an infrastructural framework – Italian Job, 1969; Die Hard, 1988. They were trailed by a moment era of movies that were characterized by, what's more, romanticized, the sex particular (male) "programmer" – War Games, 1983.

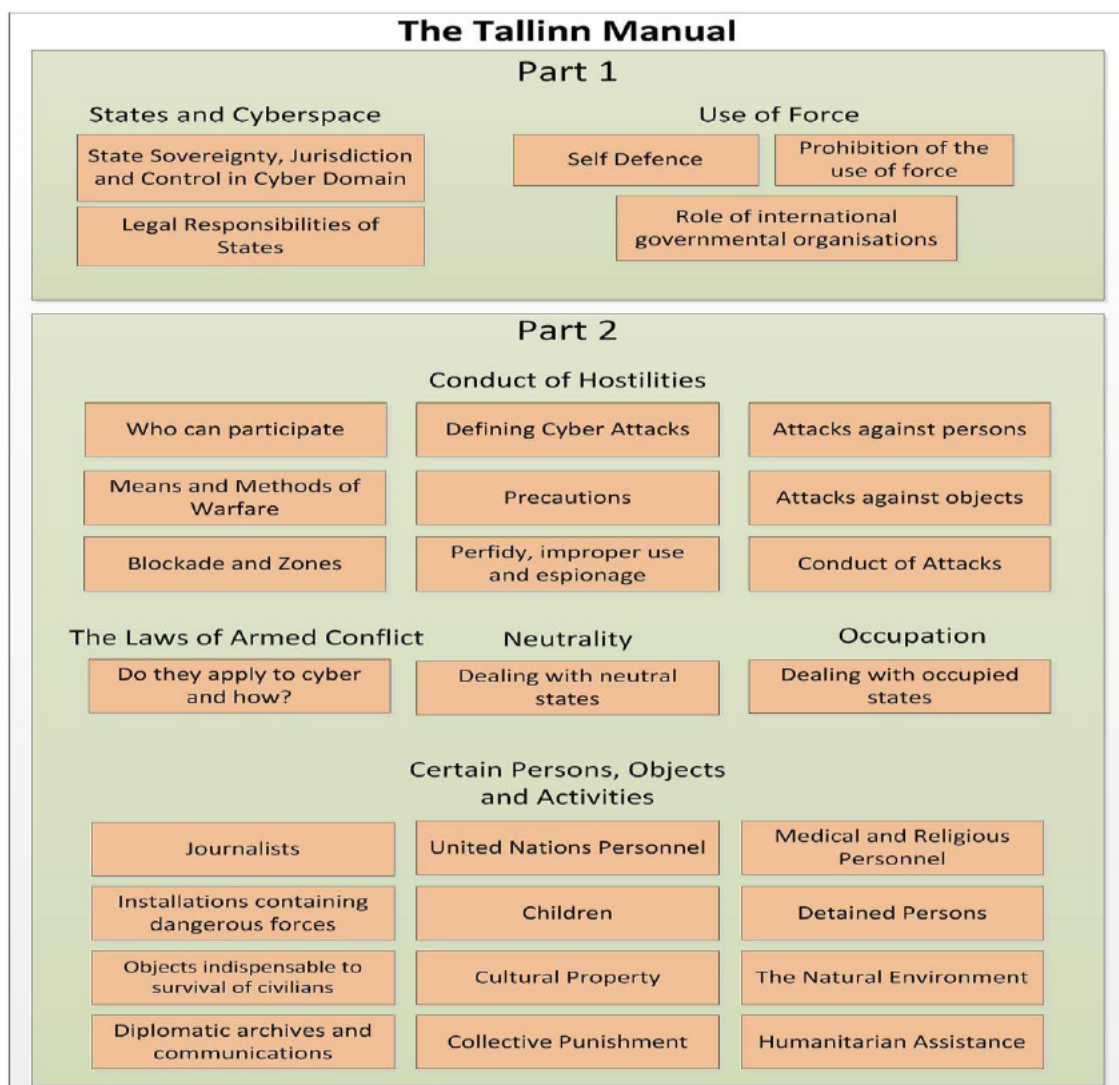
The presence of numerous theories about cybercrime – legislative–academic–expert: popular Cybercrime implies diverse things to various groups of individuals thus it is seen differently by each. Contemporary open deliberations about cybercrime incorporate a scope of legal–political, innovative, social, and monetary talks that have prompted some extremely diverse epistemological developments of cybercrime ("No Title," 2014; Svensson, 2015). The legal–administrative talk investigates what should occur by building up and elucidating the standards that recognize limits of adequate and unsuitable conduct; the criminological and general scholastic talk principally gives an educated investigation into what has happened,

and why; the security master recognizes what is really happening and what, from their own viewpoint, the strategic arrangements may be, furthermore, what will happen if those arrangements are not received; the popular-lay talk reflects what the individual in the city believes is going on. The security master talk is right now the most persuasive yet farsighted of the different talks since it consolidates self-collated factual data to shape expectations about what's to come (Arce, 2015).

Cybercrimes have a tendency to be missed by the Criminal Justice radar. This oversight presents various difficulties for the criminal equity framework to defeat, resolve or in a few conditions acknowledge. Of these, the under-detailing of cybercrimes (those which are esteemed genuine by casualties) is the main issue. Regardless of the presence of relevant collections of law moved down by worldwide harmonization and police co-appointment bargains, the attributes of cybercrimes that are grave enough to get announced, plot to obstruct the customary investigative process (Maathuis, Pieters, & Van Den Berg, 2017).

2.2. Cyberwarfare

The term cyber warfare is one that is used in mainstream media and as with information warfare, there are many differing definitions. In 2001, Alford (2001) defined cyber warfare as: “Any act intended to compel an opponent to fulfil our national will, executed against the software controlling processes within an opponent’s system.”



This definition from Alford reflects the view that cyber warfare is something that states will engage in to advance a national agenda. It can be argued, however, that modern warfare does not always aim to advance such an agenda. Religious beliefs and ideologies that are not tied to a national agenda can arguably be the aim of modern warfare. It, therefore, seems unwise to confine a definition of cyber warfare to have the purpose of advancing a national will. Jeffrey Carr (2012) offers another definition of cyber warfare:

“Cyber warfare is the art and science of fighting without fighting; of defeating an opponent without spilling their blood.” In comparison to Alford's (2001), this definition avoids attempting to define the motivation of the fighting parties. However, the suggestion that cyber warfare will not spill blood must be questioned. A cyber-attack on critical national infrastructure, such as the power grid may result in loss of life. Colarik and Janczewski (2011) agree with this point, arguing that cyber warfare cannot be seen as bloodless (Maskun, Manuputty, Noor, & Sumardi, 2013).

An examination of the writing has exhibited that there is no broadly acknowledged meaning of cyber-crime. Some researchers offer extremely wide definitions, which complete tend to cover most possible instances of cyber-crime yet are conceivably excessively wide (Onyeji, Bazilian, & Bronk, 2014). Others give particular definitions, which are potentially more valuable yet then neglect to cover certain components of what could be thought about cyber-crime. Definitions from different sources, for example, the Oxford English Dictionary have additionally been appeared to be dangerous. With the use of the term expanding in political and media circles (Vlahos, 2014; Wilking, 2013), the absence of a deliberately achieved definition is an issue that should be tended to. To determine this issue, we propose a definition demonstrate that depends on the recognizable proof of performing artists and purpose. NATO has published the Tallinn Manual (NATO, 2013), which provides some guidance on perfidy in cyber warfare. The manual states that combatants are not obliged to mark websites, IP addresses or other information technology facilities that are used for military purposes.

However, making such entities appear to have civilian status with a view to deceiving the enemy in order to kill or injure is perfidious. Secondly, it states that while concealing the origin of an attack is not perfidious, inviting the enemy to conclude that the originator is a protected person would count as perfidy. Finally, the manual concludes that conducting cyber-attacks through civilian infrastructure does not automatically make them perfidious unless it is specifically protected infra-structure such as medical systems (Lilienthal & Ahmad, 2015; Prasad, 2012; Sinopoli, 2012).

Rowe (2010) provides a military perspective on the ethics of cyberwar. He discusses issues such as ensuring civilians are made aware of what it may mean to partake in a cyber war (becoming a legal combatant, for example) and tackles the question on if fighting wars remotely can be considered ethical. He concludes by stating that cyber troops will not require physical courage, but a moral courage to do what is right without much guidance from established ethical guidelines. Dipert (2010) agrees with this view, stating that cyber warfare “appears to be almost entirely unaddressed by the traditional morality and laws of war” (Dipert, 2010).

2.3. Conducting Cyber Warfare

When a new domain of war arises, there is an immediate challenge in determining how to operate inside of it effectively. The arrival of air as a domain of war was met with research on how its properties could be leveraged to most effectively fight in it. The same process applies to the arrival of the cyber domain. This research challenge is therefore concerned with

addressing how to conduct cyber warfare, and how properties of the cyber domain shape that conduct. Parks and Duggan (2011) have examined the established principles of kinetic warfare, as defined by the US Department of Defense (Peter, 2017; Pipyros, Mitrou, Gritzalis, & Apostolopoulos, 2016; Robinson, Jones, & Janicke, 2015). They then suggest eight new principles that shape the conduct of cyber warfare. These new principles are as follows:

- Lack of physical limitations
- Kinetic effects
- Stealth
- Mutability and inconsistency
- Identity and privileges
- Dual use
- Infrastructure control
- Information as operational environment

Table 1 Libicki's seven forms of information warfare

Form	Description
Command-and-control	Attacks on command centres, or commanders themselves to disrupt command effectiveness
Intelligence-based	Increasing your own situational awareness while reducing your opponent's
Electronic	Use of cryptography and degrading the physical basis for transferring information (e.g. radar jamming)
Psychological	Use of information against the human mind. Propaganda to demoralise troops or influence civilian populations
Hacker	Exploitation of viruses, logic bombs and trojan horses to attack computer systems
Economic information	Possessing and being in control of information leads to power
Cyber	Information crime, semantic attack, simula-warfare, Gibson-warfare

Roderic Broadhurst, (2006) Cyber-crime is a regularly conventional felony (e.g. misrepresentation, robbery, child explicit entertainment) but executed quickly and to tremendous quantities of potential assaults and additionally unapproved get to, harm and obstruct the PC software. Most negative are malignant and misuse codes that intrude on PC operations on a worldwide scale and alongside different cyber-crimes debilitate e-commerce (Feiler, 2011). The cross-national way of most PC-related felonies has rendered numerous time-framed techniques for policing both locally and in cross-border circumstances insufficient even in cutting-edge countries, while the "advanced gap" gives "safe space" for cyber-criminals. Because of the risk of a cyber felony, there is an earnest need to change techniques for MLA and to create worldwide policing ability. Vatis, (2001), paper looks at contextual analyses of political clashes that have prompted assaults on digital frameworks, for example, the current conflicts amongst India and Pakistan, Israel and the Palestinians, and NATO and Serbia in Kosovo, and the pressures between the U.S. Furthermore, China over the crash between a Chinese military aircraft and an American reconnaissance plane.

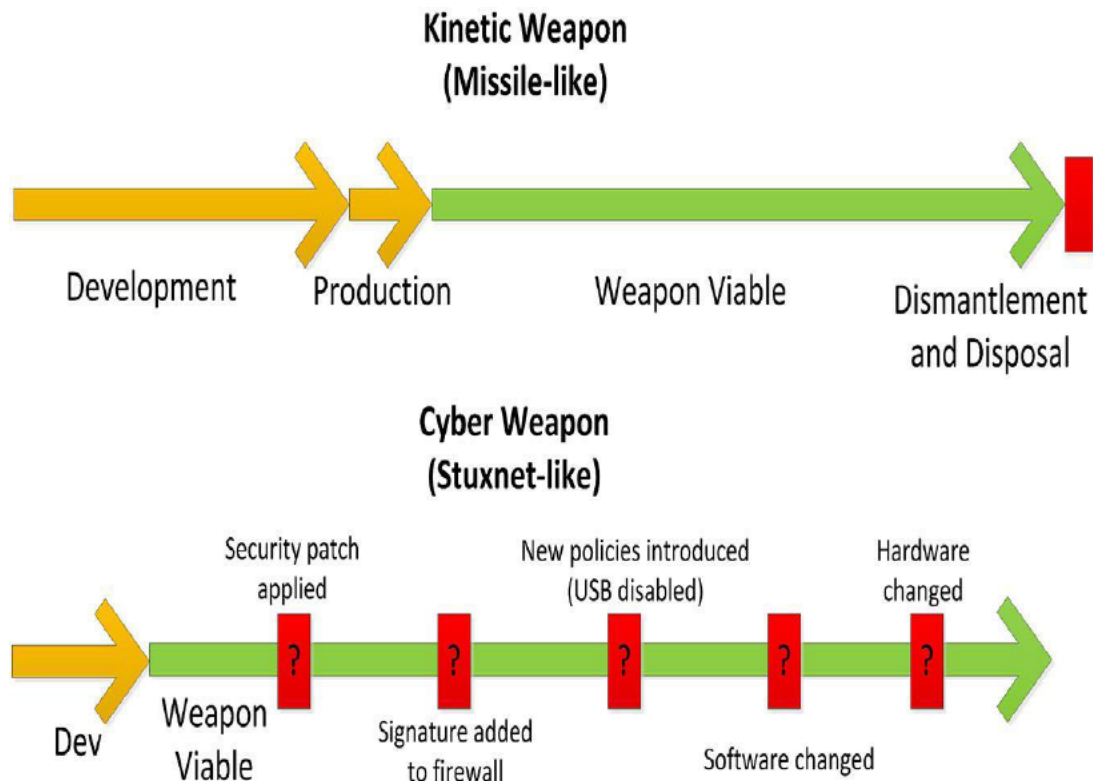


Figure 2 Cyber weapon five step process

DS Wall, (2015) while there seems, by all accounts, to be a typical view that the Internet has had a noteworthy effect upon culpability, there is substantially less accord in the matter of what that effect has been. Many sources make guarantees about the predominance of cybercrimes (organized PC felony) without clearing up what is definitely the current issue(Fellow, 2015; Jonkeren, Azzini, Galbusera, Ntalampiras, & Giannopoulos, 2015). To be sure, when alleged instances of cybercrime come to court they frequently have the recognizable ring of the "conventional" as opposed to the "digital" about them. Extortion, explicit entertainment, paedophilia, and so forth, are as of now secured by substantive zones of law in many purviews. Significantly all the more befuddling is the hole between the huge number of thousands of assessed episodes and the moderately modest number of known indictments - which addresses the early expectations that cybercrimes unless checked, could successfully bring offenders into each home. Ft Ngo & R Paternoster, (2011) Utilizing an example of understudies, the paper applies the general hypothesis of a felony and the way of life/schedule exercises system to survey the impacts of individual and situational considers on seven sorts of cybercrime exploitation. The outcomes show that neither individual nor situational qualities reliably affected the probability of being defrauded on the internet. Restraint was altogether identified with just two of the seven kinds of cybercrime exploitations and albeit five of the coefficients in the standard action models were critical, everything except one of these noteworthy impacts was in the inverse heading to that normal from the hypothesis. In any event, doubtlessly other hypothetical systems ought to speak to keeping in mind the end goal to clarify exploitation on the internet(Holder, Khurana, Hook, Bacon, & Day, 2016; Schmitt, 2014; Schmitt & NATO Cooperative Cyber Defence Centre of Excellence., 2013).

2.4. From 'Cyberterrorism' to Cyberwar', back and forth

Threats and risks from cyberspace have become a popular theme in the mass media, the security policy community, and the risk consultant and IT industry. Probably everybody with

some interest in computers and politics has already heard of the hackers, cyber terrorists or foreign intelligence agencies electronically breaking into computers that control dams or air traffic control systems, wreaking havoc and endangering not only thousands of lives but national security itself. A good indicator of the popularity of these scenarios is the fact that the entertainment sector has already capitalized on this, producing movies like the 1995 James Bond *Goldeneye* or novels like Tom Clancy and Steve R. Pieczenik's *Netforce* (Clancy and Pieczenik, 1999). Sometimes it is hard to tell what is Science and what is fiction. The whole debate started around 1990 when the Soviet Union had collapsed and the threat estimation professionals in the security community were looking for new ideas. The rise of widespread Internet use and the debate on the emerging 'information society' sparked several studies on the risks of the highly networked, high-tech-dependent United States. The National Academy of Sciences as early as 1990 began a report on computer security with the words: 'We are at risk. Increasingly,

Nowadays most of the countries depend on computers, tomorrow's terrorist may be able to do more damage with a keyboard than with a bomb' (National Academy of Sciences 1991: 7). At the same time, the prototypical term 'electronic Pearl Harbor' was coined, which linked the whole debate to a historical trauma of US warfare. It did not take very long until the highly technical problems of IT security encryption, password protection or intrusion detection systems, to name but a few – were framed in terms before known only in military policy statements. 'Password security' became 'computer security', then 'information systems security', 'information protection', and then 'information warfare' (Dornheim 1998: 60). Some people have likened cyberthreats to weapons of mass destruction. When asked by the US Senate in 1996 to compare the magnitude of the cyberthreat with nuclear, chemical or biological weapons, then CIA director John Deutch answered, 'It is very, very close to the top' ('Cyberspace attacks...' 1996). Others, like Deutch's successor George Tenet, used the term 'weapons of mass disruption' (Tenet 1997), suggesting that in the information age an electronic attack might be as dangerous to the fabric of society as an attack with intercontinental missiles would have been in the nuclear age.

3. PURPOSE OF THE STUDY

The purpose of the study is to find out the awareness among different age group about cybercrime. And also to find out the how the respondents think that such a menace could be effaced.

The objectives of the study are as follows:-

- Does familiarity with cyber-crime anything to do with the age of the respondent
- Does familiarity with cyber-crime anything to do with the education of the respondent
- Is there any association between computer usage and computer damage due to cybercrime

4. RESEARCH DESIGN AND METHODOLOGY

For this research paper consider a poll overview strategy or (questionnaire survey method) has been received to analyzing the awareness of cybercrime and designing a relevant framework with respect to cyber warfare. The method is utilized broadly in various worldwide research papers as talked about above in the literature review. Respondents were made a request to rate the chose properties/elements influencing analyzing the awareness of cybercrime and designing a relevant framework with respect to cyber warfare utilizing a Likert scale 1 to 5 (Dixit, Pandey, Mandal, & Bansal, 2017; Pandey, Dixit, Bansal, Saproo, & Mandal, 2017; Sao, Singh, Dixit, Pandey, & Singh, 2017), the respondent was represents the different type of

professionals working or consulting cyber and judicial sector through a fair and holistic approach to provide equal chance to all the stakeholders involve in the Indian construction industry. Heterogeneity in the example review was kept up by moving toward the gatherings of those respondents, those were speaking to the imperative parts of the Indian digital industry.

The study follows both the exploratory research design and Descriptive research design. Exploratory Research is administered used to study different associations with differentials of cyber-crime. The research was carried through employing the tools of Chi-Square Test and Fisher's exact test. Descriptive Research has been used to assess other information which does not fit into any explanatory/ exploratory research. And it is best to describe them as that serves the purpose best. In order to conduct market research successfully, a sample of 325 respondents was taken to know respondents' expectation from government and their awareness about cybercrime. The area of the research/population size is throughout India basis. The Primary data for the purpose of the research was collected from the customers through a structured questionnaire. A total of 325 questionnaires were taken from respondents for the research throughout the country. The Secondary data for the research was sourced from Government website and reports, magazines and journals etc.

5. CONDUCT OF RESEARCH

A structured questionnaire is floated with 350 number of respondents working in Indian IT sector and 325 valid responses were received out of 750 with a response rate of 43%. The respondents were asked to rate their responses over the Likert scale rating from 1 to 5. The questions were designed in such a way that they were simple and can easily understand the respondents.

The questionnaire is divided into two main parts are as follow:

- Overview of the researcher and the study
- The questionnaire itself

The received responses by the respondents were summarized in an Excel data sheet and the data analyzed using SPSS software and the tests and analysis are performed were discussed below.

The method used to analyze the results of the questionnaire the research was carried through employing the tools of Chi-Square Test and Fisher's exact test. For the questionnaire, the respondents were asked to rate the factors affecting construction productivity on a Likert scale of 1 to 5 as per their degree of impact and influence with respect to they observed to impacting construction productivity in the Indian context. The value given to the Likert scale rating is defined below:

- No effect (or No opinion)
- Less effect (No or minimal effect)
- Minor effect (Minor problem)
- Serious effect (Medium problem)
- Very serious effect (major problem)

It is noticed that the numbers allow to the reactions (1-5) did not show that the interims between the scales are equivalent, nor do they demonstrate outright amounts(Dixit, Mandal, Sawhney, & Singh, 2017).

6. RELIABILITY ANALYSIS

At the point when factors have been evacuated, it is essential to cross check if figure examination measured what was wanted to be measured i.e. the properties in each component framed all in all clarify a similar measure inside target measurements.(Feiler, 2011; Maskun et al., 2013; Urquhart & McAuley, 2018) In the event that traits genuinely frame the variable recognised, it is comprehended that they should sensibly relate with each other however not the ideal relationship, however. By computing Pearson relationship utilising SPSS we can evaluate the degree of connection among different factors. The estimation of $C\alpha$ for all properties computed is 0.783 which is thought to be great(Feiler, 2011; Jackson, 2016; Maskun et al., 2013; Schmitt, 2014; Urquhart & McAuley, 2018).

7. ANALYSIS AND DISCUSSION

Table 2 Chi-Square test between age of the respondents and their familiarity with Cyber Crime

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	16.349 ^a	15	.359
Likelihood Ratio	16.688	15	.338
Linear-by-Linear Association	.314	1	.575
N of Valid Cases	325		

Interpretation: Given the significance level (0.359) there stood 36 % chances of cyber-crime awareness not dependent on age. Thus we can conclude that there is no relationship between age and knowing about cyber-crime.

Table 3 Chi-Square test between education of the respondents and their familiarity with Cyber Crime

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	10.265 ^a	18	.923
Likelihood Ratio	10.789	18	.903
Linear-by-Linear Association	2.354	1	.125
N of Valid Cases	325		

Interpretation: Given the significance level (0.923) there stood 92 % chances of cyber-crime awareness not dependent on education level attained. Thus we can conclude that there is no relationship between education level and knowing about cyber-crime.

Table 4 Chi- Square test between computer usage and computer damage due to cyber crime

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	.177 ^a	3	.981
Likelihood Ratio	.177	3	.981
Linear-by-Linear Association	.026	1	.872
N of Valid Cases	325		

Interpretation: Given the significance level (0.981) there stood 98 % chances that the computer damage due to cybercrime has nothing to do with the duration of computer usage. Thus we can conclude that there is no relationship between cybercrime and the duration of computer usage.

8. FINDINGS

Cyber-crime has engulfed every aspect of the human race. But the most vulnerable are the government and financial institutions. With enormous confidential and fiscal data at their disposal, these two institutions are at most risk of being attacked. Personal users have much on a stake, given the contemporary zeal to go online for each and every activity, their privacy and money are at highest risk (Jonkeren et al., 2015; Maathuis et al., 2017).

More vulnerable are the activities which are entwined with every day like work of an individual, followed by online banking and explicit web browsing and internet data exchange. Knowing of cyber-crime is irrespective of either age or education. Thus it can be conveniently assumed that there are ample avenues through which people get acquainted with the societal happenings (Onyeji et al., 2014; Schmitt & NATO Cooperative Cyber Defence Centre of Excellence., 2013). This paper has provided a survey of contemporary thought on the challenges presented by cyber warfare. It began by looking at existing definitions of cyber war and cyber warfare and found two problems that needed resolving. Firstly, it was found that there is no widely accepted definition of either cyber war or cyber warfare. This is problematic since, without an agreed definition, it is difficult to discuss the deeper issues or even recognize when cyber warfare is occurring.

9. CONCLUSIONS

It can be concluded that cyber-crime is a menace to the society. And in the wake of Digital India, there are ample chances that such a crime would proliferate. Since the crime functions irrespective of age and education level a strategy must be charted to create mass awareness.

There is no association as such between familiarity with cyber-crime and the age of the respondent. Similarly, there is no association between familiarity with cyber-crime and the education of the respondent. That means the awareness programmes need to be communicated through different media at each and every corner. It is astonishing to see that computer damage due to cyber-crime has nothing to do with anti-virus and duration of computer usage. That says that consumers must be aware of their online presence and remain prudent while doing online activities. In contemporary times when identities are being stolen, there are no gender dimensions to it (Jackson, 2016; Maskun et al., 2013; Roscini, 2013; Von Solms & Van Niekerk, 2013).

10. RECOMMENDATIONS

- While progress is being made to address these challenges, there are still significant gaps in research that need addressing, a number of which have been identified in this paper. The most significant conclusion to be made is that the majority of challenges presented by cyber warfare cannot be solved from the perspective of just one discipline.
- Ensure the elimination of safe havens for cybercriminals, and coordinate cooperation in the investigation and prosecution of cybercrime
- Exchange information for fighting cybercrime and train, and equip law-enforcement personnel to address cybercrime.

- Protect the security of data and computer systems from cybercrime, and permit the preservation of and quick access to electronic data pertaining to particular criminal investigations;
- Ensure mutual assistance regimes for the timely investigation of cybercrime and the timely gathering and exchange of evidence, and remind the general public of the requirement to prevent and combat cybercrime.
- Design information technologies to help to prevent and detect cybercrime, and take into account both the protection of individual freedoms and privacy and the preservation of the capacity of Governments to fight cybercrime.
- The way to form a viable responsive system to fight cybercrime is to comprehend the diverse points of view that the diverse people in the field of cybercrime convey to the subject, instead of seeing them in twofold terms as either right or wrong.
- It is likewise vital to hold reasonable desires of what the police can and can't do. This incorporates tolerating that not all policing lies with the police on the grounds that the policing capacity is likewise to be found in other nodal and organized structures of the request.
- In the perfect world, the administration of on the web conduct ought to hence be intended to help and reinforce the Internet's regular slant to police itself, keeping levels of intercession applicable while introducing suitable structures of responsibility.

11. LIMITATIONS OF THE STUDY

The quality of this paper is its exhaustive examination and positioning of traits analyzing the awareness of cybercrime and designing a relevant framework with respect to cyber warfare, in which the statistic, geographic, designing and financial condition might be not quite the same as that in which a few different investigations have been led. Its significant shortcoming is that the discoveries require extra inside and out research and approval. Keeping in mind the end goal to build up this exploration further, a focus examination of the vital discoveries of the review with the professionals is already in progress. This study is required for both approve the discoveries of this exploration and prompt the improvement of potential systems to limit the impact of those components that have been evaluated by this study to have the best potential analyzing the awareness of cybercrime and designing a relevant framework with respect to cyber warfare(Hobe, 2016; Lilienthal & Ahmad, 2015; Robinson et al., 2015).

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